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Reducing porosity and quality enhancement in keyhole laser welding under 3D configurable external magnetic field stimulation

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ABSTRACT

The application of external static magnetic fields during laser welding is known to reduce defect rates, improve melt-pool stability and affect the microstructure of the materials processed. The penetration of the field through the entire volume of the process zone makes it a promising method to improve welding quality but it involves complex interactions between magnetic fields and different states of matter that are absent in classical welding. However, most of today's research is focused on applications of static magnetic fields that are well elaborated, while alternating fields remain "terra-incognito" and their potential has still not been fully characterized. Fortunately, the generation of suitable alternating fields has recently become possible with the availability of appropriate commercial hardware. This study introduces a novel approach for reducing porosity in keyhole-mode laser welding using a 3D-configurable external magnetic field system capable of spatial–temporal modulation. While prior research primarily focuses on static magnetic fields, this work explores the effects of alternating (AMF) and rotating magnetic fields (RMF) with tunable direction and frequency—up to 200 Hz—on pore dynamics. Laser welding experiments on AlSi10Mg and 316L stainless steel revealed porosity reductions of up to 71 % under Bxy-200 Hz conditions. Unlike existing methods, our system dynamically responds to melt pool instabilities, enabling real-time force control via Lorentz and eddy current effects. CT analysis confirmed significant pore collapse and redistribution. Additionally, elemental mapping and metallographic analysis suggest enhanced alloy mixing and melt back-filling driven by electromagnetic stirring. We also identify a novel pore destabilization mechanism under AMF/RMF exposure, not attributed solely to Marangoni or Lorentz forces. These findings position configurable magnetic fields as a transformative tool for defect mitigation in advanced manufacturing.

1. Introduction

Laser welding (LW) stands as a pivotal technique in modern manufacturing, delivering unmatched precision and deep penetration of welds that is essential for joining critical components in key economic sectors such as aerospace [1], automotive [2], naval [3] and medical device production [4]. The growing importance of this technology also relates to several additional advantages, such as narrow area of the heat affected zone, low energy consumption, high processing speed at low running costs, possibility to process complex material combinations, better material fusion in the process zone etc. [5,6]. For these reasons LW is replacing traditional arc welding in a growing number of industrial applications [6]. However, despite obvious advantages, LW is prone to defect formation that are a hidden threat for the mechanical properties of the processed parts [7]. This is especially relevant for LW in the keyhole regime, where light energy provokes material vaporization, thereby forming deep and narrow vapor cavities known as keyholes. Keyholes enhance the weld penetration depth and improve material fusion, however its inherent instability introduces defects such as porosity, weld beads and spatter [8]. Porosity forms due to the trapping of parts of the keyhole channel by the melt pool, resulting in empty voids after processing. Pores weaken mechanical strength and fatigue resistance [8]. The irregular weld beads and spatter are characterized by material rejection from the melt pool, compromising structural

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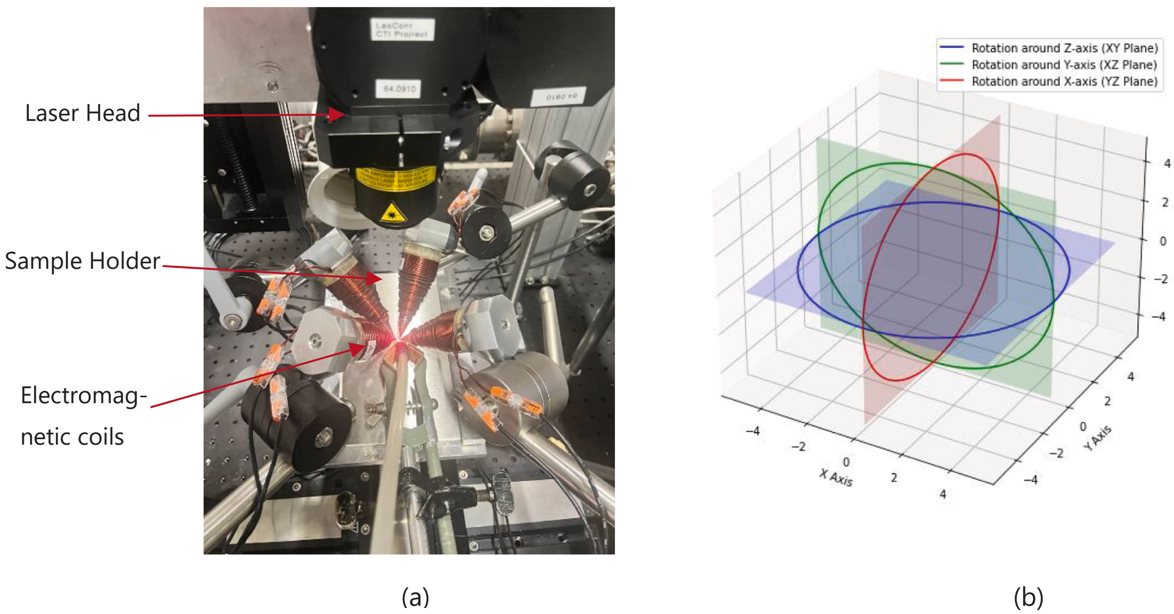

Fig. 1. (a) Experimental setup with laser and electromagnetic coils, and (b) 3D circular planes representing rotation magnetic field modulation.

reliability [8,9]. These complex interactions, driven by non-stationary temperature-induced mass flows, alternating temperature gradients, and varying solidification dynamics [10], make the minimization of defect rates challenging.

Efforts to address these issues have spanned various strategies, such as in-process control for stabilizing the keyhole [2,3]. However, the process dynamics make this approach challenging and it has not gone beyond the laboratory stage [10]. Simpler approaches include the use of shielding gases [1]. The use of argon suppresses oxidation and stabilizes the melt pool to some extent, though it offers limited influence over internal flow dynamics [1]. While these methods can improve the weld quality, they lack the precision to fully mitigate the complex interplay of forces driving defect formation.

LW in external magnetic fields (here and further referred to as M-fields) have emerged as a transformative approach to enhance keyhole laser welding [11]. By interacting with the electrically conductive molten metal, M-fields induce eddy currents that generate Lorentz forces, that affect mass flows, suppress turbulences and, consequently, redistribute thermal gradients [12,13]. Studies have demonstrated their potential: static M-fields have reduced porosity, deepened weld penetration, and refined grain structures by leveraging electromagnetic braking (known as the Hartmann effect) to dampen instabilities [14,15]. Time-varying M-fields further enhance these benefits, promoting alloy mixing and enabling defect-free welds in complex geometries [16,17]. For instance, research by Meng et al. [14] showed that magnetic stirring improves metal mixing, while Xu et al. [16] reported uniform microstructures under static field support. These findings underscore the usefulness of M-fields as a powerful tool for weld quality improvement. Pioneering work by Kern et al. [18] showed that a 40 mT M-field during CO2 laser welding smoothed weld surfaces, driven by thermoelectric currents from the Seebeck effect. Ambrosy et al. [19] found stronger M-field effects in CO2 (10.6 μm wavelength) versus solid-state laser welding (1.06 μm), linked to higher plasma-induced currents. Zohm et al. [20] modeled thermoelectric currents, confirming measured potential differences. Chen et al. [17] showed that Lorentz forces (1.0×10^6 N/m^3) extend the columnar crystal region, with currents (1.0×10^7 A/m^2) concentrated at the molten pool boundary. Bachmann et al. [9,21] demonstrated that steady M-fields reduce flow velocity via the Hartmann effect, improving weld bead morphology, while AC fields (230 mT RMS, 2.6 kHz) counteract sagging in thick joints. However, fixed keyhole models in these studies overlooked oscillations and thermocapillary convection, which Pang et al. [22,23] identified as critical to molten pool dynamics.

Despite reported successes in the aforementioned works a critical limitation persists: most studies rely on static or pre-configured M-field setups with fixed parameters—orientation, strength, and frequency—optimized for specific conditions [12,14]. This rigidity fails to account for the non-stationary nature of welding, where thermal conditions, material properties, and phase boundaries evolve rapidly, rendering static fields inadequate for real-world variability [12]. The inability to dynamically configure M-fields limits their adaptability, leaving a gap in understanding how field variations influence keyhole stability and pore removal mechanisms across diverse scenarios.

In this report we introduce a novel approach to bridge this gap: a 3D configurational magnetic modulation system that dynamically adjusts M-field parameters—orientation, strength, and frequency—in real-time. Unlike static setups, our system exploits the proportionality between Lorentz forces and M-field characteristics [12] to respond instantaneously to melt pool fluctuations, offering unprecedented control over weld dynamics. This study focuses on unravelling the mechanisms behind keyhole stability and porosity reduction under varied M-field configurations, from static to rotating and alternating fields. We hypothesize that eddy currents, looping around non-conductive gas pores, locally reduce surface tension, destabilizing pore interfaces and promoting collapse, while dynamic Lorentz forces enhance gas expulsion and back-filling. By systematically exploring these effects, our work not only addresses the current control limitations and prediction capabilities but also provides a mechanistic framework to optimize M-field-assisted laser welding, paving the way for adaptive, defect-free manufacturing solutions.

This paper consists of four sections: i) Introduction, ii) Experimental Setup, iii) Results and iv) Discussion. In the Conclusion section, we have proposed detailed future perspectives and an outlook.

2. Experimental setup

2.1. Laser welding with magnetic field stimulation setup

As shown in Fig. 1, the setup incorporates a magnetic field generator developed by Magnebotix AG (Switzerland), consisting of four fixed

Table 1
Material properties for AlSi10Mg and 316L.

	316L	AlSi10Mg
Electrical Conductivity (S/m)	6×10^5	3.5×10^5
Thermal Conductivity (W/m·K)	15	120–150
Melt Viscosity (mPa·s)	6	3–4

electromagnets positioned 4 mm from the process zone, the primary region of interest. The alignment of the system is performed using a laser pointer directed onto the sample, ensuring precise positioning of the magnetic field relative to the weld track.

The magnetic field system was calibrated using a Hall-effect 3D magnetic sensor (Infineon TLV493D-A1B6, tolerance 0.1 mT) positioned at the workpiece surface. Iterative current adjustments ensured precise field strengths of 170 mT in X, 190 mT in Y, and 100 mT in Z. Three magnetic field configurations were investigated: Static Magnetic Fields, Alternating Magnetic Fields (AMF), and Rotating M-Fields (RMF). These were applied in the X, Y, and Z directions, as well as in the XY, XZ, and YZ planes, respectively. For AMF and RMF conditions, frequencies of 50 Hz, 100 Hz, and 200 Hz were tested, with 200 Hz being the upper operational limit of the system. The 200 Hz system limit restricts exploration of higher-frequency effects, where reduced skin depth could further concentrate Lorentz forces. Future studies should investigate frequencies above 200 Hz to enhance melt pool stabilization. Field configurations were denoted accordingly: static fields as Bx Static, By Static, etc.; AMF fields as Bn followed by the frequency (e.g., Bx 50 Hz for an alternating field in the X-direction at 50 Hz); and RMF fields as Bnm followed by the frequency (e.g., Bxy 50 Hz for a rotating field in the XY-plane at 50 Hz), where 'n' represents the direction for AMF and 'nm' the plane for RMF. Relative to the laser system, the X-direction corresponds to the transverse direction of the weld track, the Y-direction aligns with the weld path (scanning direction), and the Z-direction is perpendicular to the substrate surface.

The laser processing was performed using a StarFiber150 P1 laser (Coherent, Switzerland), operating at a wavelength of 1070 nm in both pulsed and continuous-wave modes. The laser beam was delivered through a single-mode optical fiber (12 μm core diameter) and directed to a custom optical head (Coherent, Switzerland), which focused the beam to a 43 μm spot size Don the surface of the workpiece. All processing was conducted in an argon atmosphere to prevent oxidation and ensure process stability. The argon shielding gas was supplied through a fixed nozzle positioned 5 mm from the laser spot at a 45° angle relative to the workpiece surface, ensuring effective protection against oxidation. The argon flow rate was maintained at 21 L/min to stabilize the melt pool and minimize contamination. The workpieces were mounted on an M-663.5U precision translation stage (Zaber, Canada), enabling controlled relative motion between the optical head and the sample for high-precision surface scanning. The processing parameters for the weld tracks are as follows: for 316L, the laser power was 100 W with a stage speed of 3 mm/s; for AlSi10Mg, the laser power was 250 W with the same stage speed of 3 mm/s.

This study examined two materials—AlSi10Mg and 316L steel—selected due to their contrasting properties, which are listed in Table 1:

All of which critically govern electromagnetic field interactions and porosity mitigation efficacy. The 2 mm thick 316L stainless steel sheets were procured from McMaster-Carr (USA), whereas the 4 mm thick AlSi10Mg sheets unavailable as prefabricated material due to the alloy's cast nature were additively manufactured. The AlSi10Mg samples were fabricated using a SISMA Mysint DMLS (Direct Metal Laser Sintering) system under the following process parameters: laser power of 175 W, scanning speed of 1275 mm/s, hatch distance of 12 μm, and layer thickness of 30 μm. The oxygen content in the printing environment was maintained below 100 ppm, ensuring minimal oxidation. The optical density of the printed samples was measured at 99.3 %, indicating high densification. Post-printing, the AlSi10Mg samples were milled to a final thickness of 3 mm. To ensure a uniform surface finish, all samples were polished using 1200-grit Silicon carbide (SiC) abrasive paper and subsequently sandblasted to remove surface irregularities and prepare them for further processing.

All processed samples were subjected to the porosity analysis of the weld tracks conducted using X-ray computed tomography (CT) with a V-Tome-X S240 system equipped with a 240 kV reflection tube. High-resolution imaging was achieved with a voxel size of 10.9 μm, enabling the detection of pores ≥30 μm in diameter. CT analysis (detection limit ≥30 μm) may underestimate total porosity by missing smaller pores. Future micro-CT analysis could capture finer defects. The X-ray source was operated at 195 kV and 56 μA, with a 0.3 mm copper prefilter to improve contrast. Each projection was captured with an integration time of 600 ms and averaged over three acquisitions to enhance image quality. Although the total weld track length was 25 mm, only the central 8 mm region was scanned and analyzed to minimize edge effects and focus on a representative area of the track. The porosity reduction ratio between the experimental (magnetic field applied) and reference (no/without field) conditions was calculated using the ratio of reference porosity volume to experimental porosity volume.

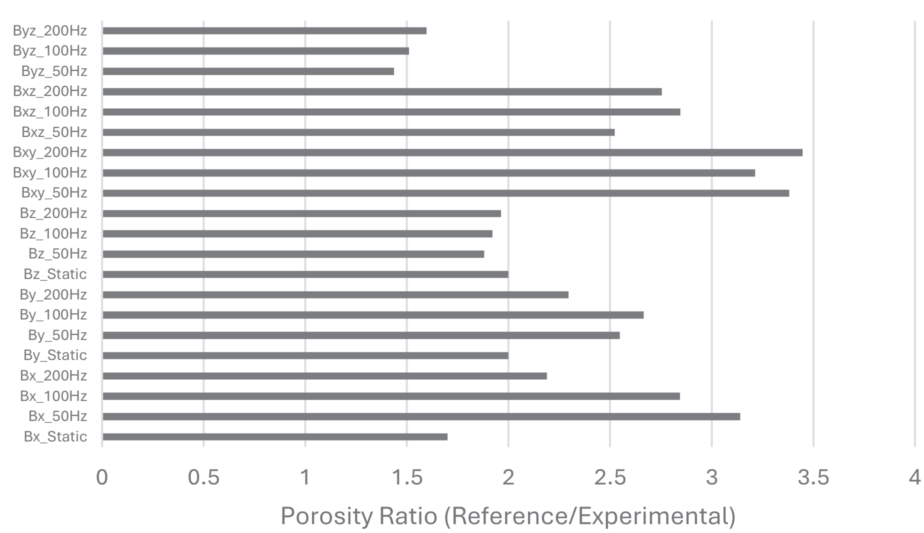

Fig. 2. Porosity volume ratio under different experimental conditions for AlSi10Mg. A ratio greater than 1 indicates a reduction in porosity relative to the reference (without magnetic field).

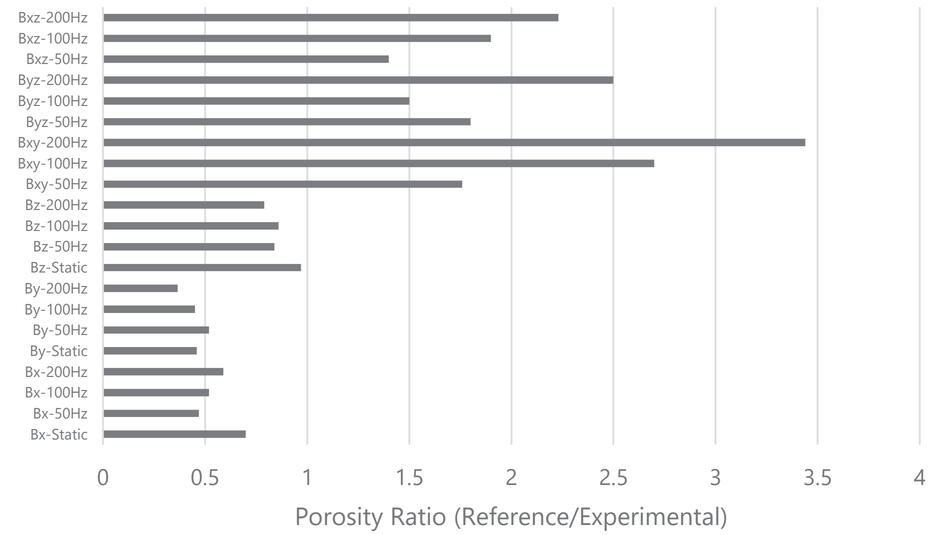

Fig. 3. Porosity volume ratio under different experimental conditions for 316L. A ratio greater than 1 indicates a reduction in porosity relative to the reference (without magnetic field).

$$PorosityRatio = \frac{ReferencePorosityVolume}{ExperimentalConditionVolume}$$

A ratio greater than 1 indicates improved suppression of defects due to the magnetic field, quantifying its effect on melt pool dynamics. To ensure statistical reliability, results were averaged across three weld tracks for each condition. Error bars are not included for porosity volume ratios, as they were calculated from the total volume of three samples, providing a single representative ratio for each condition.

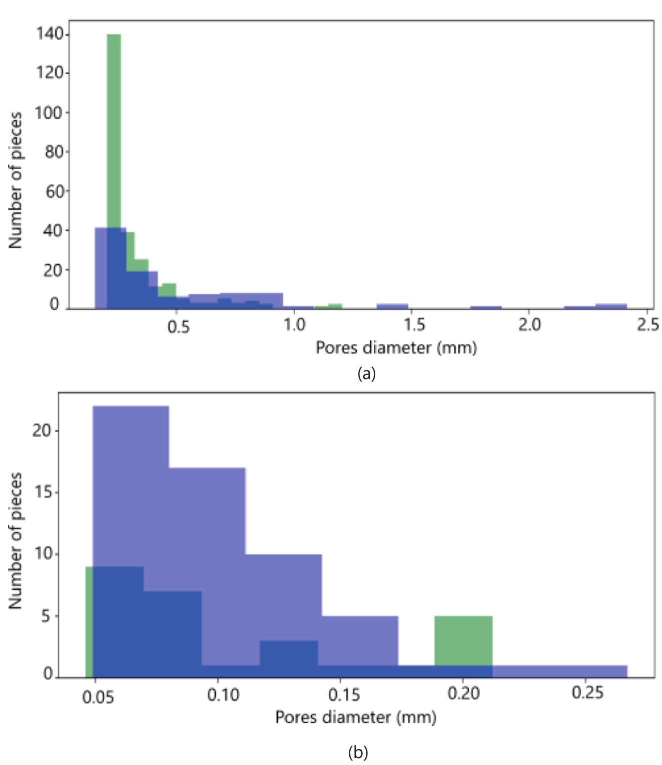

Fig. 4. Comparison of pore size distribution for (a) AlSi10Mg: No Field vs. Bxy-200 Hz, and (b) 316L: No Field vs. Bxy-200 Hz. *Note: Reference condition is shown in blue; experimental condition is shown in green.* (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3. Results

Fig. 2 illustrates the ratio (Reference (No Magnetic Field) Volume/ Experimental Volume) of porosity volumes for all experimental conditions for AlSi10Mg as quantified by CT. The influence of AMF and RMF on the porosity of AlSi10Mg, analyzed via CT, revealed substantial reductions in porosity volume, with performance strongly dependent on field orientation and frequency. In the absence of a magnetic field, the baseline porosity volume measured 1.18 mm^3. Application of electromagnetic fields reduced porosity by up to 71 %, with the most effective configurations being rotating magnetic fields in the XY plane at 200 Hz (Bxy-200 Hz: 0.34 mm^3). Rotating fields (Bxy, Bxz) consistently outperformed unidirectional AM fields, suggesting that multidirectional Lorentz forces enhance fluid flow dynamics, enabling more efficient pore dispersion and back-filling [1,2].

Fields applied perpendicular to the scanning direction (X) achieved superior porosity reduction compared to those aligned with the weld path (Y). For instance, Bx-50 Hz reduced porosity by 68 %, whereas By-50 Hz yielded only a 61 % reduction. This anisotropy aligns with prior studies [3], where transverse Lorentz forces (orthogonal to the welding direction) stabilize the keyhole by suppressing vapor cavity oscillations, thereby minimizing process pores. Conversely, longitudinal fields (parallel to the welding direction) exhibited weaker influence, likely due to limited interaction with keyhole instabilities. Rotating fields (e.g., Bxy-200 Hz) further amplified porosity reduction, likely by inducing turbulent multidirectional flows that disrupt pore nucleation and enhance gas expulsion through vigorous melt pool agitation [4].

The superior performance of rotating fields, particularly Bxy-200 Hz, underscores the role of azimuthal Lorentz forces in generating convective flows capable of redistributing gas bubbles. These forces likely promote pore back-filling by driving bubbles toward the melt pool surface or dispersing them into smaller, less detrimental voids [2]. The 71 % porosity reduction under Bxy-200 Hz aligns with electromagnetic stirring mechanisms reported in laser welding of aluminum alloys [5], where similar field configurations reduced defect volumes by destabilizing pore-forming regimes.

Notably, metallurgical pores near the weld root persisted across all conditions, albeit reduced in volume. This suggests that while electromagnetic fields effectively mitigate process pores (e.g., keyhole-driven voids), residual porosity may arise from gas entrapment during solidification, which is less sensitive to Lorentz forces [6]. The results highlight the efficacy of rotating magnetic fields in optimizing melt pool dynamics, particularly when multidirectional agitation disrupts keyhole

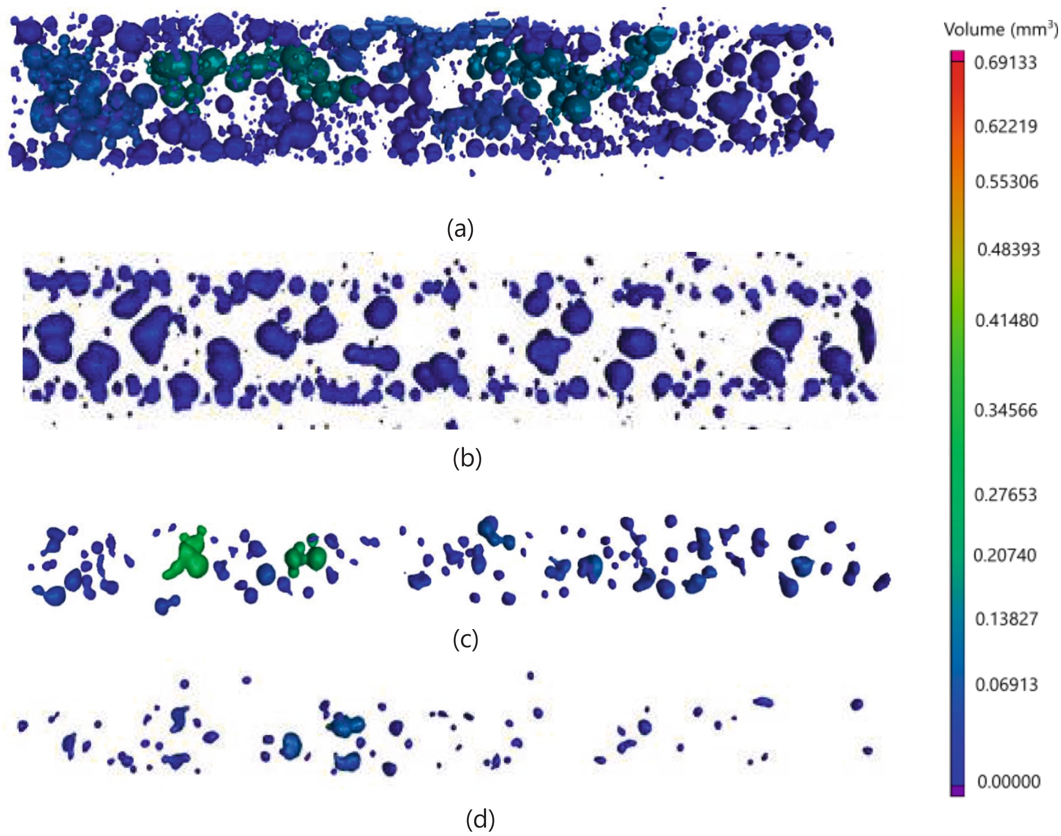

Fig. 5. CT reconstruction images showing pore distribution: (a), (b) AlSi10Mg without and with Bxy-200 Hz; (c), (d) 316L without and with Bxy-200 Hz. All images show central 8 mm region.

instabilities or enhances back-filling.

Similarly, the porosity volume ratio in 316L stainless steel, quantified via CT, exhibited significant variations depending on the electromagnetic field configuration and frequency is shown in Fig. 3. The reference porosity volume (without magnetic field) was 0.13 mm^3, while experimental conditions ranged from 0.03 mm^3 (Bxy-200 Hz) to 0.36 mm^3 (By-200 Hz), reflecting reductions or increases of up to 70.9 % and 173 %, respectively. Rotating magnetic fields (RMF) in the XY plane demonstrated superior performance: Bxy-200 Hz achieved the lowest porosity volume (0.03 mm^3, 70.9 % reduction), followed by Bxy-100 Hz (0.04 mm^3, 63.0 % reduction) and Bxy-10 Hz (0.07 mm^3, 43.2 % reduction). This frequency-dependent efficacy underscores the critical role of dynamic Lorentz forces in suppressing porosity, as higher-frequency rotating fields generated stronger azimuthal agitation to disrupt pore nucleation and enhance back-filling.

Unidirectional AMF yielded mixed results. Transverse fields perpendicular to the welding direction (e.g., Bz-200 Hz, 0.10 mm^3, 20.1 % reduction) modestly reduced porosity, while longitudinal fields aligned with the weld path (By-200 Hz, 0.36 mm^3, 173 % increase) exacerbated pore formation, likely due to destabilization of the keyhole. Porosity increased by 173 % (0.36 mm^3 vs. 0.13 mm^3) at By-200 Hz in 316L due to excessive keyhole agitation from unidirectional magnetic fields misaligned with the welding direction, destabilizing the melt pool. Static magnetic fields universally underperformed, with porosity volumes exceeding the reference (e.g., Bx-static: 0.18 mm^3, 43.0 % increase), highlighting the inability of steady Lorentz forces to perturb melt pool dynamics meaningfully.

A comparison of pore size distributions between the reference (no magnetic field) and Bxy-200 Hz conditions for both AlSi10Mg and 316L stainless steel reveals a distinct shift toward smaller pore diameters

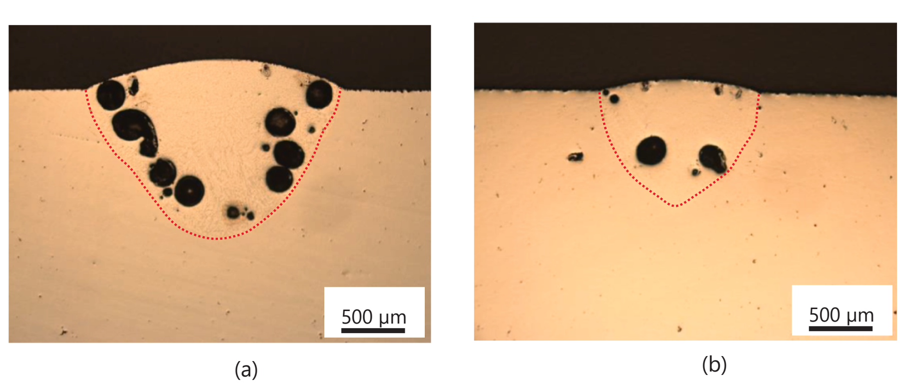

Fig. 6. Optical micrographs showing the melt pool geometry in AlSi10Mg samples: (a) without magnetic field, and (b) under Bxy-200 Hz.

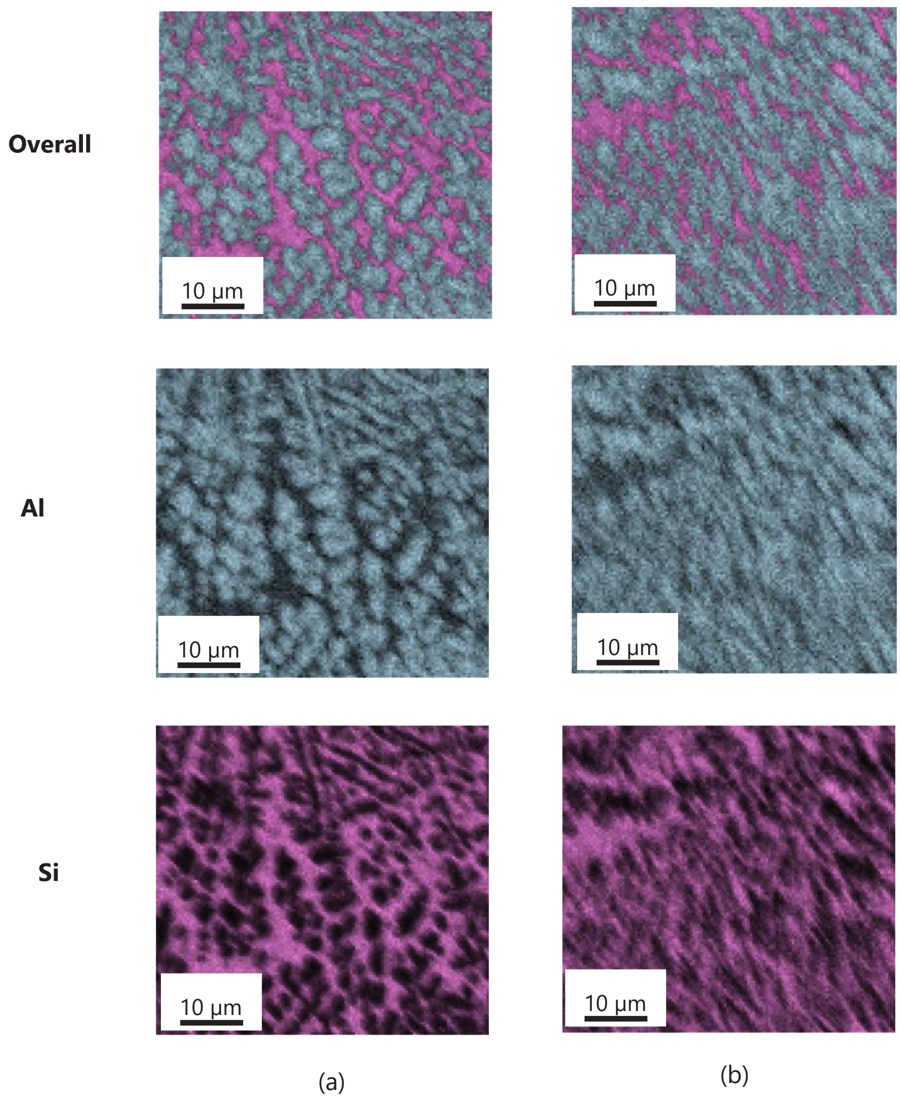

Fig. 7. EDX elemental mapping showing refined Si-rich phases in AlSi10Mg under: (a) without magnetic field, and (b) under Bxy-200 Hz, highlighting electromagnetic stirring effects.

under the influence of the rotating magnetic field, as shown in Fig. 4. This observation supports the hypothesis that eddy currents generated by the electromagnetic field disrupt the surface tension at the gas–liquid interface, promoting an accelerated back-filling mechanism. It is also possible that larger pores undergo fragmentation into smaller voids as an intermediate step in their eventual elimination. The same trend is reflected in the CT reconstructions presented in Fig. 5, where a notable reduction in both pore size and overall porosity can be observed under Bxy-200 Hz. The effectiveness of this back-filling process is closely related to the melt viscosity of the material, which influences the dynamics of fluid flow and pore closure. This relationship will be explored further in the discussion section.

We observed a noticeable influence of the rotating magnetic field on the melt pool geometry in AlSi10Mg samples. In the absence of an external magnetic field, the melt pool exhibited a width of 1.65 ± 0.2 mm and a depth of 1.16 ± 0.1 mm. However, under the influence of a 200 Hz rotating magnetic field in the XY plane (Bxy-200 Hz), the melt pool dimensions were significantly reduced, with a width of 1.03 ± 0.1 mm and a depth of 0.87 ± 0.1 mm. These observations suggest that the applied magnetic field leads to a more confined and possibly stabilized melt pool. The corresponding melt pool shapes for both cases are shown in Fig. 6. In contrast, 316L samples did not exhibit a comparable change in melt pool morphology under the same magnetic field conditions. The calculated Hartmann numbers for AlSi10Mg and 316L stainless steel are approximately 4.99 and 2.66, respectively. These values were determined using the applied magnetic field strength of 190 mT, melt pool widths of 1.6 mm (AlSi10Mg) and 1.4 mm (316L), electrical conductivities of 3.5×10^5 S/m (AlSi10Mg) and 6.0×10^5 S/m (316L), and dynamic viscosities of 1.3×10^{-3} Pa·s and 6.0×10^{-3} Pa·s, respectively. The higher Hartmann number in AlSi10Mg indicates a stronger electromagnetic influence on the melt pool, consistent with the observed reduction in melt pool width and depth under magnetic field application. Conversely, the lower Hartmann number for 316L corresponds to the minimal change in melt pool morphology. However, while the Hartmann number provides insight into the magnetohydrodynamic behavior of the melt pool, porosity reduction mechanisms involve additional factors such as electromagnetic stirring, melt viscosity, and pore back-filling, which are discussed in detail in the following section.

It is also evident that the application of a rotating magnetic field induces electromagnetic stirring within the melt pool, which enhances the mixing of alloying elements during solidification. This effect was confirmed through energy-dispersive X-ray spectroscopy (EDX) elemental mapping of Al and Si in samples processed without a magnetic field and with Bxy-200 Hz. In the absence of a magnetic field, larger Si-

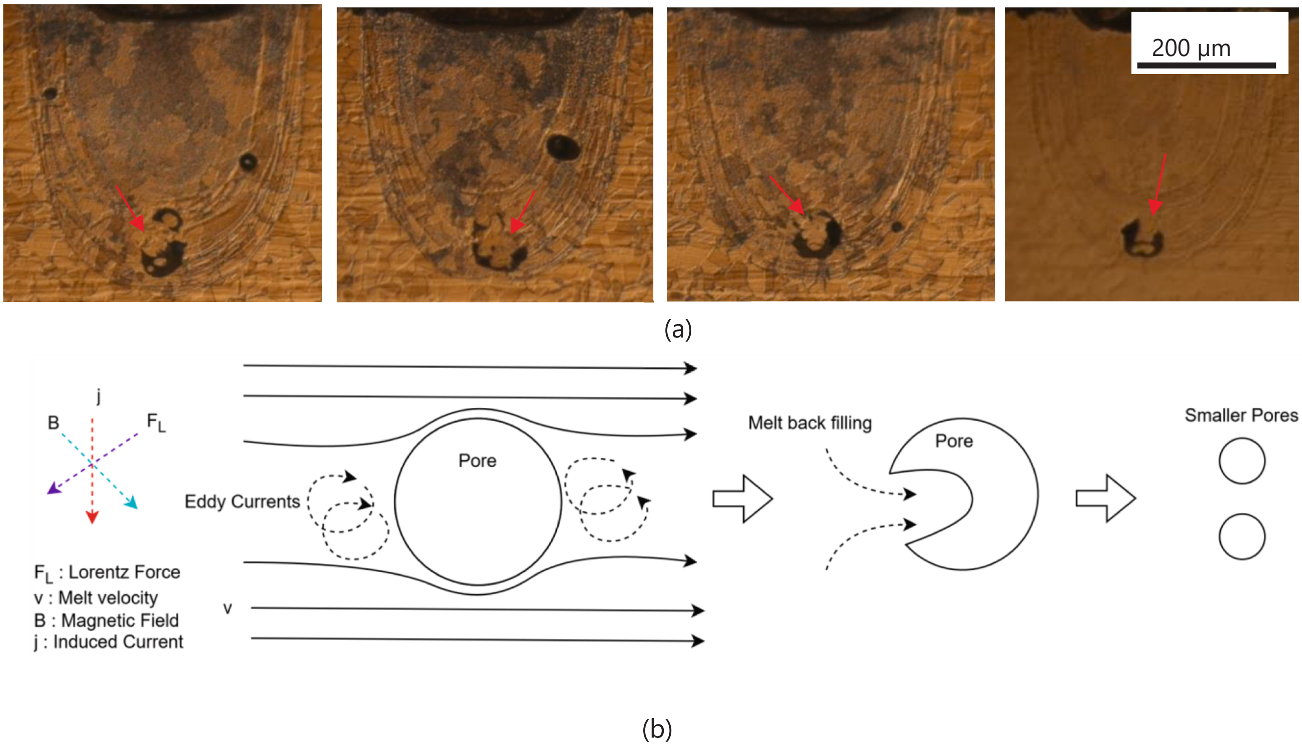

Fig. 8. (a) Optical micrograph of 316L showing partially filled (half-filled) pores (marked by red arrow), indicating ongoing pore back-filling during solidification. (b) Schematic illustration of the proposed mechanism, highlighting pore fragmentation, electromagnetic-driven fluid flow, and subsequent back-filling contributing to pore elimination. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

rich regions were observed, indicating localized segregation. In contrast, under Bxy-200 Hz conditions, the Si-rich phases appeared finer and more uniformly distributed. This refinement is attributed to the influence of electromagnetic stirring on the solidification process, promoting dendrite fragmentation and a more homogeneous microstructure, as illustrated in the corresponding Fig. 7.

4. Discussion

Beyond the established mechanisms, microscopy analysis revealed a previously unreported pathway of pore mitigation that occurs specifically under time-varying magnetic fields—namely AMF and RMF magnetic fields—but is absent under static field conditions. As observed in Fig. 8a, in 316L stainless steel, certain pores appear to be collapsing inward due to wall failure, leaving behind partially filled voids. This phenomenon suggests a process distinct from mere convective melt flow.

We propose that this collapse results from the interplay between eddy currents, localized Joule heating, and the thermodynamic destabilization of gas pores. Under AMF or RMF exposure, time-varying magnetic fields induce circulating eddy currents (J_e) in the conductive molten metal surrounding each gas pore. However, these currents cannot penetrate the non-conductive gas phase, and instead concentrate around the pore boundaries. The resulting localized Joule heating ($Q = \frac{J_e^2}{\sigma}$, $\sim 10^3\ W/m^3$, for 316L) raises the temperature specifically at the pore–melt interface. Due to 316L's relatively low thermal conductivity, this heat does not dissipate quickly, forming steep local temperature gradients.

This localized temperature rise lowers the surface tension (γ) at the pore boundary—consistent with temperature-dependent surface energy behavior in liquid metal [14,24]. As γ decreases, the Laplace pressure ($\Delta P = \frac{2\gamma}{r}$) inside the pore also drops, weakening the internal gas pressure that stabilizes the pore structure. Consequently, the surrounding molten metal can overcome this diminished resistance, driving material into the pore cavity.

Simultaneously, the oscillating Lorentz forces (F_L) generated by AMF induce pulsating shear stresses ($\tau\ \frac{F_L}{A}$) at the gas–liquid interface. When these shear stresses exceed the restoring surface tension forces ($\tau > \frac{\gamma}{r}$), the pore wall integrity is further compromised, promoting collapse or fragmentation [25]. For RMF, the swirling azimuthal Lorentz forces create helical convection patterns that sustain continuous melt agitation, further enhancing melt infiltration and gas displacement.

Despite these driving forces, the high viscosity (η) of 316L inhibits rapid melt flow ($v\ \frac{F}{\eta}$), and thus back-filling is often incomplete before solidification. This results in partially filled pores, a signature of incomplete melt infiltration under transient conditions. Similar observations have been made in electromagnetic-assisted laser cladding of high-viscosity alloys, where pores are disrupted but not fully closed [26].

In contrast, static magnetic fields (where $\frac{\partial B}{\partial t} = 0$) generate steady Lorentz forces that lack the oscillatory energy needed to generate eddy currents, localized heating, or time-dependent shear stresses. As a result, the melt experiences only unidirectional flow, insufficient to disturb surface tension or promote back-filling. The gas pores in this case remain largely unaffected, which aligns with findings in studies of solidification under static magnetic fields [27].

Furthermore, transient Joule heating under AMF/RMF conditions momentarily reduces local viscosity in 316L, enhancing fluidity at the pore interface. However, the alloy's low thermal conductivity accelerates solidification around the heated zones, often freezing the melt before complete pore filling occurs [28]. This interplay between transient viscosity reduction and rapid boundary solidification contributes to the trapping of partially collapsed or half-filled pores.

Finally, this effect is material-dependent. In AlSi10Mg, which has a significantly lower viscosity and higher thermal conductivity than 316L, the same electromagnetic mechanisms lead to more complete pore evacuation. The molten metal more easily infiltrates the destabilized pores before solidification concludes—explaining why half-filled pores are not observed in this alloy under identical field conditions [29].

In summary, the emergence of half-filled pores in 316L under AMF and RMF conditions can be attributed to eddy current-induced localized heating, which lowers surface tension and pore pressure, enabling partial collapse and melt back-filling. The incomplete nature of this back-filling in high-viscosity alloys highlights the critical role of both electromagnetic field dynamics and alloy-specific thermophysical properties in governing pore behavior. These insights are consistent with prior observations in nickel-based superalloys, where viscosity-dependent pore evolution has been reported under electromagnetic processing conditions [26].

5. Conclusion

This study demonstrates that dynamically modulated magnetic fields—particularly high-frequency rotating fields—can significantly reduce porosity in keyhole-mode laser welding of AlSi10Mg and 316L. The highest porosity reduction (71 %) was achieved under Bxy-200 Hz conditions, underscoring the importance of field orientation and frequency. We report the emergence of new pore destabilization mechanisms involving eddy current-induced heating, Lorentz-driven shear, and viscosity-modulated back-filling. These mechanisms, previously unreported in static field configurations, were particularly effective in low-viscosity alloys like AlSi10Mg. In contrast, high-viscosity alloys such as 316L exhibited partial pore filling, highlighting the material dependence of field effects. Our findings advocate for integrating 3D-configurable magnetic field systems into next-generation laser welding platforms. Future work will focus on real-time field sensing, melt pool simulation, and in situ imaging to further elucidate transient interactions during welding and enable adaptive, closed-loop control for defect-free fabrication.

CRediT authorship contribution statement

Pinku Yadav: Writing – original draft, Validation, Methodology, Investigation, Data curation, Conceptualization. **Chang Rajani:** Visualization, Investigation. **Simone Gervasoni:** Supervision, Resources, Methodology. **Enea Masina:** Data curation. **David Sargent:** Supervision, Resources, Project administration, Investigation. **Patrik Hoffmann:** Supervision. **Elia Iseli:** Project administration. **Sergey Shevchik:** Project administration, Methodology, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Sergey Shevchik reports financial support was provided by Innosuisse Swiss Innovation Agency. Pinku Yadav reports a relationship with EMPA Swiss Federal Laboratories for Materials Science and Technology that includes: employment. Sergey Schevchik has patent pending to EU Patent Office. No Conflict of Interest If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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